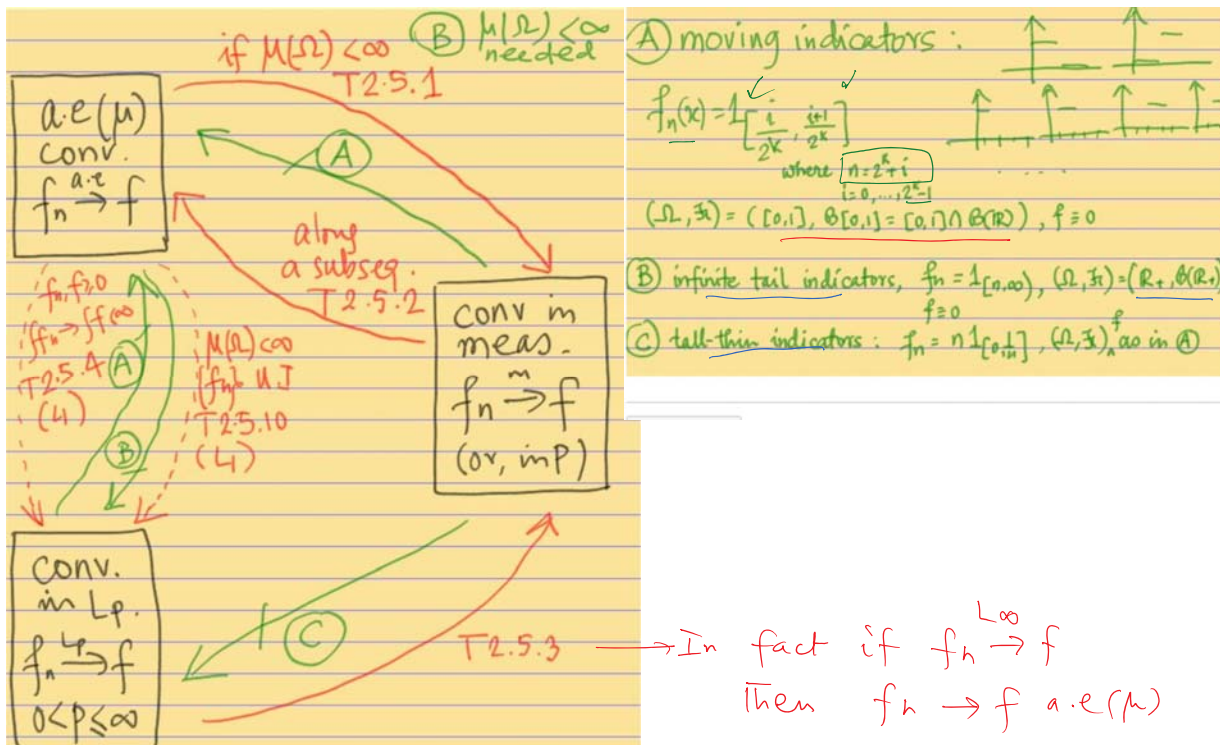
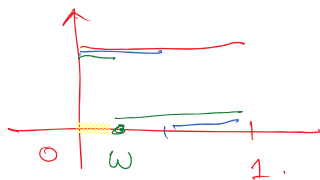




(A)



$$f_1 = 1_{[0, 1]}$$

$$n = 1 = 2^0 + 0$$

$$k = 0$$

$$f_2 = 1_{[0, \frac{1}{2})}, f_3 = 1_{[\frac{1}{2}, 1]}$$

$$k = 1$$

$$f_4 = 1_{[0, \frac{1}{4})}, f_5 = 1_{[\frac{1}{4}, \frac{1}{2})}, f_6 = 1_{[\frac{1}{2}, \frac{3}{4})}, f_7 = 1_{[\frac{3}{4}, 1]}$$

$$k = 2$$

fix $\omega \in \Omega = [0, 1]$

$f_n(\omega) = 0$ or 1 infinitely many times

$$f \equiv 0$$

$$0 < \varepsilon < 1$$

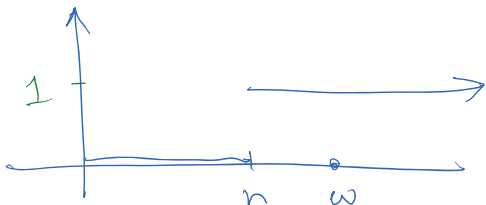
$$\mu(\{\omega : |f_n(\omega) - f(\omega)| > \varepsilon\}) = \frac{1}{2^k} \text{ if } 2^k \leq n < 2^{k+1}$$

So $f_n \xrightarrow{\text{in meas.}} f$ but $f_n \not\xrightarrow{\text{a.e.}} f$ $\frac{1}{2^k} < \frac{2}{n} \rightarrow 0$ as $n \rightarrow \infty$ $\frac{1}{2^k} = \frac{2}{2^{k+1}} < \frac{2}{n}$

(B)

$$f_n = 1_{[n, \infty)}, (\Omega, \mathcal{F}, \mu) = (\mathbb{R}_+, \mathcal{B}(\mathbb{R}_+), m)$$

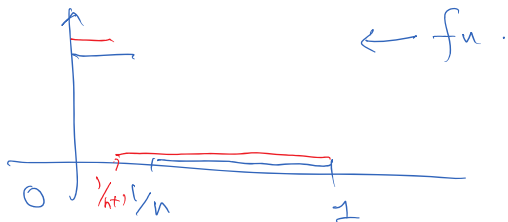
$$[0, \infty)$$



$$f \equiv 0$$

$f_n \xrightarrow{\text{a.e.}(\mu)} f$ but not in meas.

③



a.e.(n)
 $f_n \rightarrow f$ but not in L_1

* check the other statements in diagram.

take $\varepsilon > 0$. if $\varepsilon \geq 1$ then
 $\mu(|f_n - f| > \varepsilon) = 0$

if $0 < \varepsilon < 1$

$$\mu(|f_n - f| > \varepsilon) = \mu((n, \infty)) = \infty$$

$\rightarrow 0$ as $n \rightarrow \infty$.

for $\omega \in \Omega = \mathbb{R}_+$

take $N > \omega$, $f_n(\omega) = 0 \quad \forall n > N$.
 $\rightarrow f(\omega) = 0$

Thm 2.51: $\mu(\Omega) < \infty$,

given $\mu\left(\lim_{n \rightarrow \infty} f_n \neq f\right) = 0$, show $\lim_{n \rightarrow \infty} \mu(|f_n - f| > \varepsilon) = 0$

$\Updownarrow$

$$\mu\left(\limsup_{n \rightarrow \infty} |f_n - f| > 0\right) = 0$$

A.

will show $\forall r > 1$
 $\Rightarrow \lim_{n \rightarrow \infty} \mu(|f_n - f| \geq \frac{1}{r}) = 0$

take any $\varepsilon > 0$, choose
 $r > \frac{1}{\varepsilon}$ then $\frac{1}{r} \leq \varepsilon$

$$\mu(|f_n - f| > \varepsilon) \leq \mu(|f_n - f| \geq \frac{1}{r}) \rightarrow 0$$

$$A = \left\{ \omega : \limsup_{n \rightarrow \infty} |f_n(\omega) - f(\omega)| > 0 \right\}.$$

$$= \{ \omega : \limsup |f_n(\omega) - f(\omega)| \geq \frac{1}{r} \text{ for some } r \geq 1 \}$$

$$= \bigcup_{r \geq 1} \left\{ \omega : \inf_{n \geq 1} \left(\sup_{j \geq n} |f_j(\omega) - f(\omega)| \right) \geq \frac{1}{r} \right\}.$$

$$= \bigcup_{r \geq 1} \left\{ \omega : \sup_{j \geq n} |f_j(\omega) - f(\omega)| \geq \frac{1}{r}, \forall n \geq 1 \right\}$$

$$= \bigcup_{r \geq 1} \bigcap_{n \geq 1} \left\{ \omega : \sup_{j \geq n} |f_j(\omega) - f(\omega)| \geq \frac{1}{r} \right\}$$

$$= \bigcup_{r \geq 1} \left[\bigcap_{n \geq 1} \left(\bigcup_{j \geq n} \underbrace{\left\{ \omega : |f_j(\omega) - f(\omega)| \geq \frac{1}{r} \right\}}_{B_j(r)} \right) \right]$$

$$C_n(r) = \bigcup_{j \geq n} B_j(r)$$

$$D(r) = \bigcap_{n \geq 1} C_n(r)$$

$$A = \bigcup_{r \geq 1} D(r) \quad \text{and} \quad \mu(A) = 0$$

$$\mu(A) = 0 \Rightarrow \mu(D(r)) \leq \mu(A) = 0 \quad \forall r \geq 1$$

$$C_n(r) \downarrow D(r) \text{ as } n \rightarrow \infty. \text{ and } \mu(\Omega) < \infty$$

$$\text{so m.c.f.a. applies : } \lim_{n \rightarrow \infty} \mu(C_n(r)) = \mu(D(r)) = 0$$

$$\text{Note } \mu(B_j(r)) \leq \mu(C_n(r)) \quad \forall j \geq n.$$

$$\mu(B_n(r)) \leq \mu(C_n(r)) \rightarrow 0 \text{ as } n \rightarrow \infty \quad \forall r \geq 1$$

$$\lim \mu(B_n(r)) = 0 \quad \forall r \geq 1$$

$$\lim_{n \rightarrow \infty} \mu(B_n(r)) = 0 \quad \forall r \geq 1$$

i.e., $\lim_{n \rightarrow \infty} \mu\left(\left\{w : |f_n(w) - f(w)| \geq \frac{1}{r}\right\}\right) = 0$
pf complete.

Remark 2.5.1. (Problem 2.37 (a) + (b) .)

(a) real analysis fact: $x_n \rightarrow x \xLeftrightarrow \textcircled{1} \forall \{x_{n_k}\} \subset \{x_n\}$
 $\xLeftrightarrow \textcircled{2} \exists \{x_{n_{k_l}}\} \subset \{x_{n_k}\}$
 s.t. $x_{n_{k_l}} \rightarrow x$.

(b) DCT for $f_n \xrightarrow{m} f$.

$f_n \xrightarrow{m} f$ and $|f_n| \leq g$, $g \in L_1$, then

$$\int |f_n - f| d\mu \rightarrow 0$$

Pf: take any such $\{f_n\}$. will show $\underbrace{\int |f_n - f| d\mu}_{x_n} \rightarrow 0$

using ② above, choose any subseq. f_{n_k} of f

will show $\exists f_{n_{k_l}}$ s.t. $\int |f_{n_{k_l}} - f| d\mu \rightarrow 0$.

Since $f_n \xrightarrow{m} f$, $f_{n_k} \xrightarrow{m} f$ for any subseq.

for this subseq, $\exists$ sub-subseq $\{f_{n_{k_l}}\}$ s.t. $f_{n_{k_l}} \xrightarrow{a.e.} f$
 (T.2.5.2)

using DCT (or EDCI / BCT / MCT)

on this sub-subseq., we get the result.

Problem 2.21 & 2.22.

Cont version of MCT / DCT etc.

Cont version of DCT:

if $g_t \rightarrow g$ a.e. (μ) and $|g_t| \leq h$, $h \in L_1$

then $\int |g_t - g| d\mu \rightarrow 0$ as $t \rightarrow \infty$.

Pf: $g_t \rightarrow g$ a.e. (μ) $\Rightarrow$ $g_{t_n} \rightarrow g$ a.e. (μ)
usual DCT applies.

$\rightarrow \int |g_{t_n} - g| d\mu \rightarrow 0$.
Since true for any $\{t_n\}$,
we get the result.

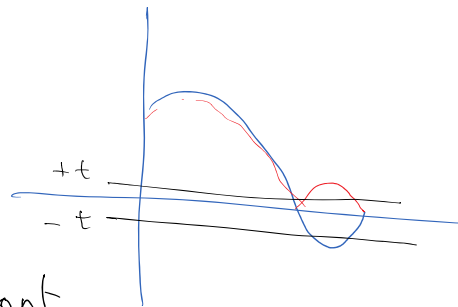
Now take $f \in L_1 \Rightarrow \int |f| d\mu < \infty \Rightarrow |f| < \infty$ a.e. (μ)

take $g_t = |f| \mathbb{1}_{\{|f| > t\}}$, $t > 0$.

$\rightarrow 0 = g$ a.e. (μ)

as $t \rightarrow \infty$

also, $|g_t| \leq |f|$.



So $\int g_t d\mu \rightarrow \int g d\mu = 0$ by cont DCT.

So if $f \in L_1$, Then $\int |f| \mathbb{1}_{\{|f| > t\}} d\mu \rightarrow 0$ as $t \rightarrow \infty$.

Similarly, $f, g \in L_1$, Then can show

$$\max \left(\int |f| \mathbb{1}_{\{|f| > t\}} d\mu, \int |g| \mathbb{1}_{\{|g| > t\}} d\mu \right) \rightarrow 0$$

here we say $\{f, g\}$ are uniformly integrable

(a property of the collection
of int. functions)

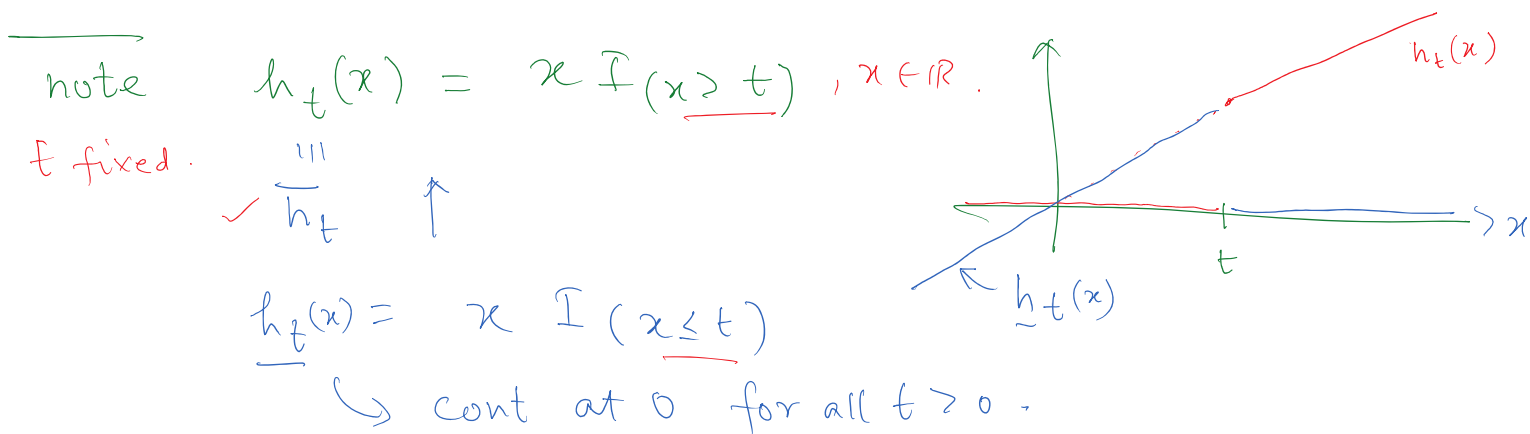
$\{f_\lambda\}_{\lambda \in \Lambda} \subseteq L_1(\Omega, \mathcal{F}, \mu)$ is u.i. if $h_t(f_\lambda)$

$$\sup_{\lambda \in \Lambda} \int |f_\lambda| \mathbb{1}_{\{|f_\lambda| > t\}} d\mu \rightarrow 0 \text{ as } t \rightarrow \infty.$$

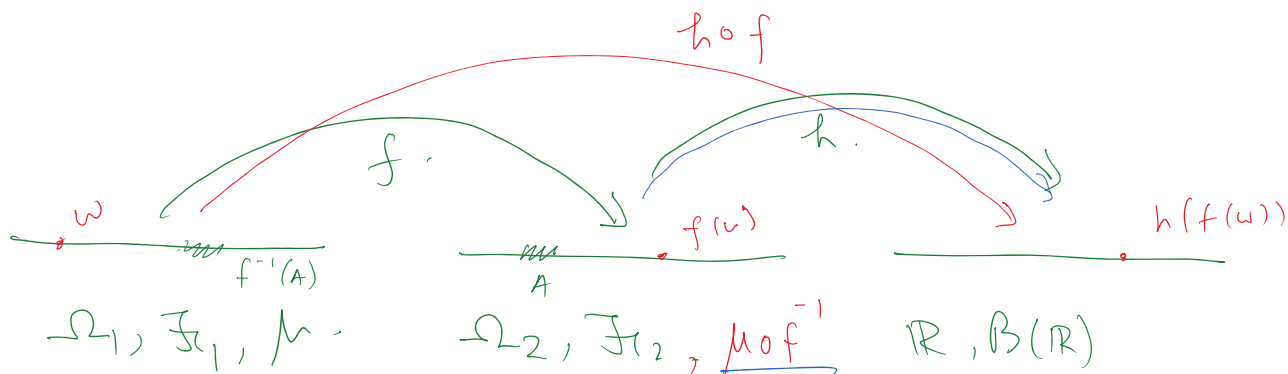
$\varphi(t)$

$\{X_n\}_{n \geq 1} \subseteq L_1(\Omega, \mathcal{F}, P)$ is u.i. if

$$\sup_{n \geq 1} E(|X_n| \mathbb{I}(|X_n| > t)) \rightarrow 0 \text{ as } t \rightarrow \infty$$



2.42 (Change of Variable).



$$(\mu \circ f^{-1})(A) = \mu(f^{-1}(A))$$

$\downarrow$
induced meas.

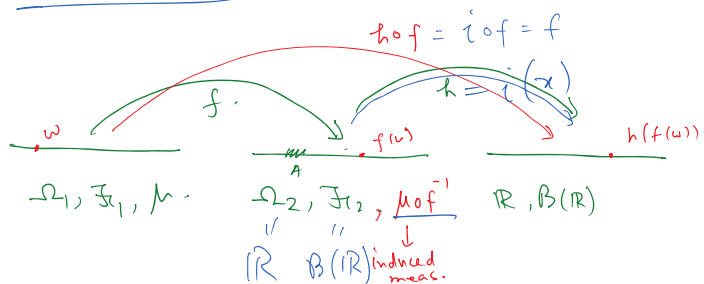
$$\int_{\Omega_1} (h \circ f) d\mu = \int_{\Omega_2} h d(\mu \circ f^{-1})$$

take $h = 1_A$,

$$\begin{aligned}
 \text{LHS} &= \int_{\Omega_1} 1_A(f(\omega)) d\mu(\omega) = \int_{\Omega_1} 1_{f^{-1}(A)}(\omega) d\mu(\omega) \\
 &= \mu(f^{-1}(A)) \\
 &= (\mu \circ f^{-1})(A) \text{ by defn.} \\
 &= \int_{\Omega_2} 1_A d(\mu \circ f^{-1}) = \text{RHS}
 \end{aligned}$$

then go through the usual steps - simple, $h \geq 0$, $h \in L$, etc.

2 versions



$$\int_{\Omega_1} f d\mu = \int_{\Omega_1} (i \circ f) d\mu = \int_{\Omega_2} i(x) d(\mu \circ f^{-1})(x)$$

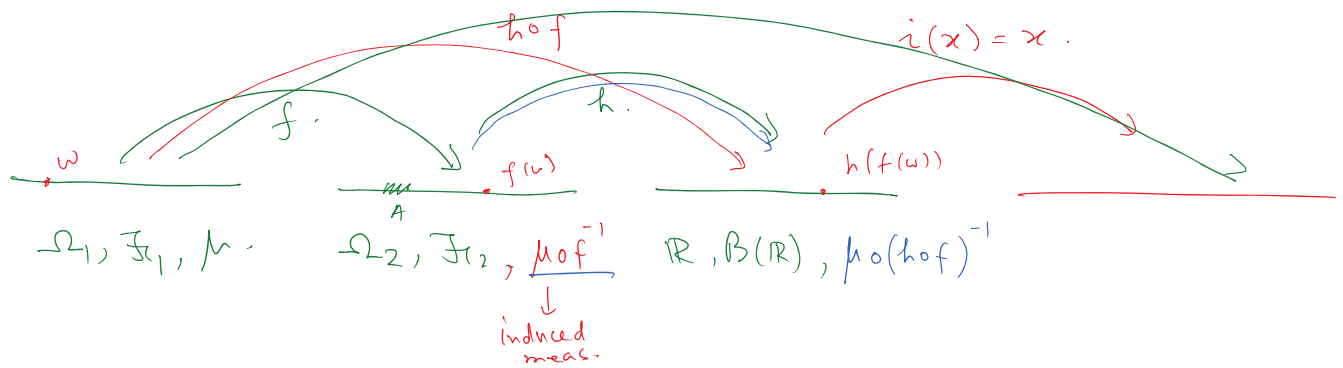
in particular if $\mu \geq P$, $f = X$

$$E(X) \equiv \int_{\Omega} X dP = \int_{\mathbb{R}} x dP_X(x)$$

distr. of r.v. X

$$\begin{aligned}
 P_X(A) &= (P \circ X^{-1})(A) \\
 &= P(X^{-1}(A)), \quad A \in \mathcal{B}(\mathbb{R})
 \end{aligned}$$

$i \circ h \circ f$



$$\int_{\Omega_1} i(h \circ f) d\mu = \int_{\Omega_2} i \circ h d(\mu \circ f^{-1}) = \int_{\Omega_3 = \mathbb{R}} i d(\mu \circ (h \circ f)^{-1})$$

ex! $X \sim \text{Bin}(n, p)$, compute $E(X)$.



$$P_X(\{i\}) = \begin{cases} \binom{n}{i} p^i (1-p)^{n-i}, & i = 0, 1, \dots, n \\ 0, & \text{otherwise.} \end{cases}$$

$P(X \geq 3) = P_X([3, \infty))$, we know $\exists (\underline{\Omega}, \mathcal{F}, \mathbb{P})$ and meas $X: (\underline{\Omega}, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ s.t. $P_X = P \circ X^{-1}$ is like this.

$$E(X) = \int_{\underline{\Omega}} X dP = \int_{\mathbb{R}} x dP_X(x)$$

$$= \int_{\mathbb{R}} \left(\sum_{i=0}^n i \mathbb{1}_{\{i\}}(x) \right) dP_X(x)$$

$$= \sum_{i=0}^n i P_X(\{i\})$$

$$= \sum_{i=0}^n i \binom{n}{i} p^i (1-p)^{n-i} = \text{usual calc.} = np.$$

note
 $i(x) = \sum_{i=0}^n i \mathbb{1}_{\{i\}}(x)$ a.e. (P_X)
 $\mathbb{R} \rightarrow \mathbb{R}$
 $i(x) = x$
 $x \mapsto x$



ex: $X \sim \text{Normal}(0, 1)$, find $E(X)$

$$P_X(A) = \int_A \phi(x) d\mu(x) \quad \xrightarrow{\text{leb.}}$$

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}, \quad -\infty < x < \infty$$

$$E(X) = \int_{\Omega} x dP = \int_{\mathbb{R}} x dP_X(x) \quad \left(= \int_{\mathbb{R}} x dP_X \right)$$

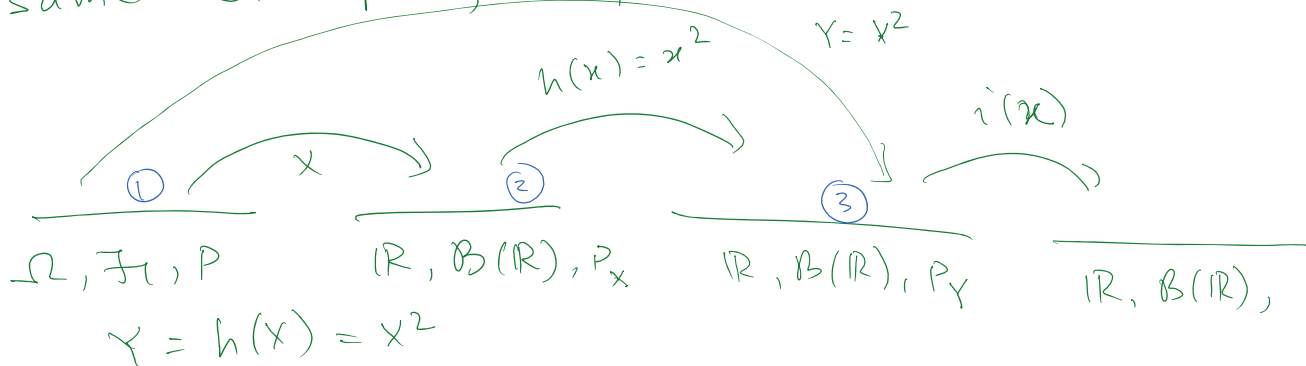
$$= \int_{\mathbb{R}} x \cdot \phi(x) d\mu(x) \quad \leftarrow \text{result earlier.}$$

$$= \dots \text{argue it is R-int.}$$

$$= \int_{-\infty}^{\infty} x \phi(x) dx \quad (\leftarrow \text{R-int.})$$

$$= 0 \quad (\text{using usual calculus})$$

ex: same example, compute $E(X^2)$



$$\int_{\Omega} Y dP \quad (1) = \int_{\mathbb{R}} h(x) dP_X(x) \quad (2) = \int_{\mathbb{R}} i(x) dP_Y(x) \quad (3)$$

$\parallel$
 $\int x^2 dP_X(x)$

$Y \sim \chi^2_{(1)}$
 $f_Y(y) = \dots$

$$\int_{\mathbb{R}} x^2 dP_X(x)$$

$$\int_{-\infty}^{\infty} x^2 \phi(x) dx$$

$$E(Y) = 1$$

Law of unconscious statistician ¹ LOTUS.